MATH 128a: Homework #13

Instructor: F. Weber

Due by: **Wednesday, December 10, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Romberg Integration

Solve problems 2ad, 12(1), 17(13), 18(14), 19(15) of Section 4.5 of the textbook.
(If the problem has a different number in the 9th edition book, it is in parentheses)

Problem 2: ODEs

Solve problems 1ad, 4bd, 6a(8a), 10(9) of Section 5.1 of the textbook.
(If the problem has a different number in the 9th edition book, it is in parentheses)

Problem 3: Euler's method

Solve problems 2, 4, 10, 12 of Section 5.2 of the textbook.
(If the problem has a different number in the 9th edition book, it is in parentheses)

(For 10c, compute a Lipschitz contant on the region $D = [1, 2] \times [-1.05, -0.45]$ and argue that the proof of Theorem 5.9 remains valid for sufficiently small h when D is a rectangle $[a, b] \times [c, d]$ instead of a strip $[a, b] \times \mathbb{R}$ provided that the exact solution and the numerical solution stay within the rectangle. Show that this is the case here.)

Problem 4: Runge-Kutta methods

Solve problems 2, 6, 14, 29(27), 30, 32 of Section 5.4 of the textbook. You may want to write a matlab function for some of the numerical methods to compute the approximations.
(If the problem has a different number in the 9th edition book, it is in parentheses)

(For 30, the equation should be $y_{i+1} = y_i + ha_1 f(t_i, y_i) + ha_2 f(t_i + h\alpha_2, y_i + h\delta_2 f(t_i, y_i))$ with the notation from the lectures. For simplicity, consider the autonomous case $f(t, y) = g(y)$. It suffices to consider just $f(t, y) = y$, but it is more instructive to leave it in the more general form $f(t, y) = g(y)$.)